

Data Preprocessing Pipeline — Primer dataset (frontonly_conf60)

(1) Raw video frame



frame dari
rekaman webcam

(2) Face detection



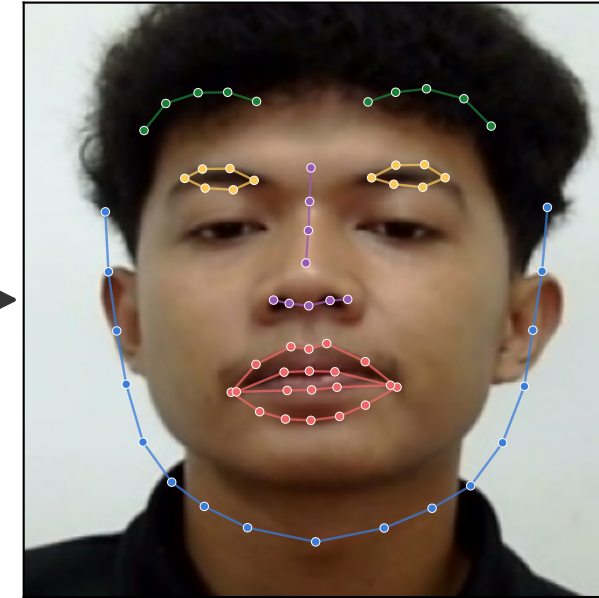
MediaPipe FaceLandmarker
bounding box

(3) Crop 224x224



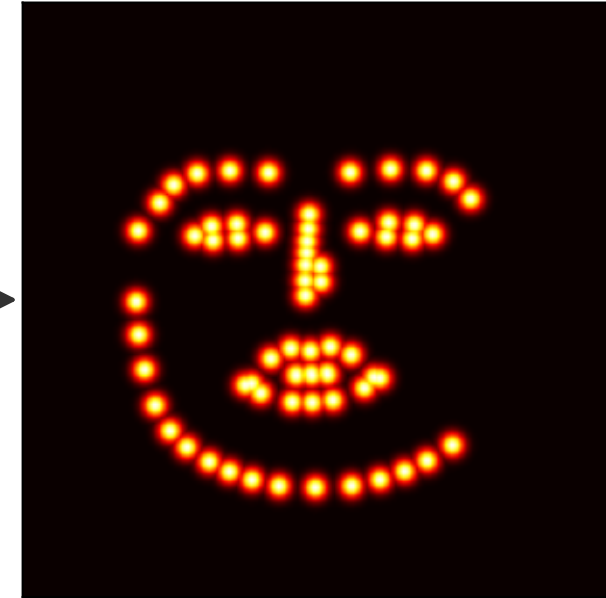
face crop
resize ke 224x224

(4) Landmark extraction



68 titik face-api.js (FA)
(MP di-extract paralel)

(5) Heatmap rendering



Gaussian blob
per landmark → channel-4

OUTPUT yang disimpan ke disk per sample:

- RGB image (224x224x3) → input CNN / image branch
- Landmark raw_136 (68x2) — 2 source paralel: face-api.js (FA) + MediaPipe (MP)
→ derive ke FACS_28 (Euclidean dist), Blendshape_52 (MP only), atau FACS+BS_80
- Heatmap (224x224) → channel ke-4 input Early Fusion